SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

Course Name	Cloud Computing and Big Data Analytics
Course Code	CSA1513
Course Outcome Covered	CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)
Dave's Taxonomy	Manipulation (Level 2)
Question Count	5
Total Marks	50
Duration	60 Minutes

Code of Conduct

I K.V.yathish Chandra(192524323) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.Yathish

Assessment Tasks – Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weight	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: _____

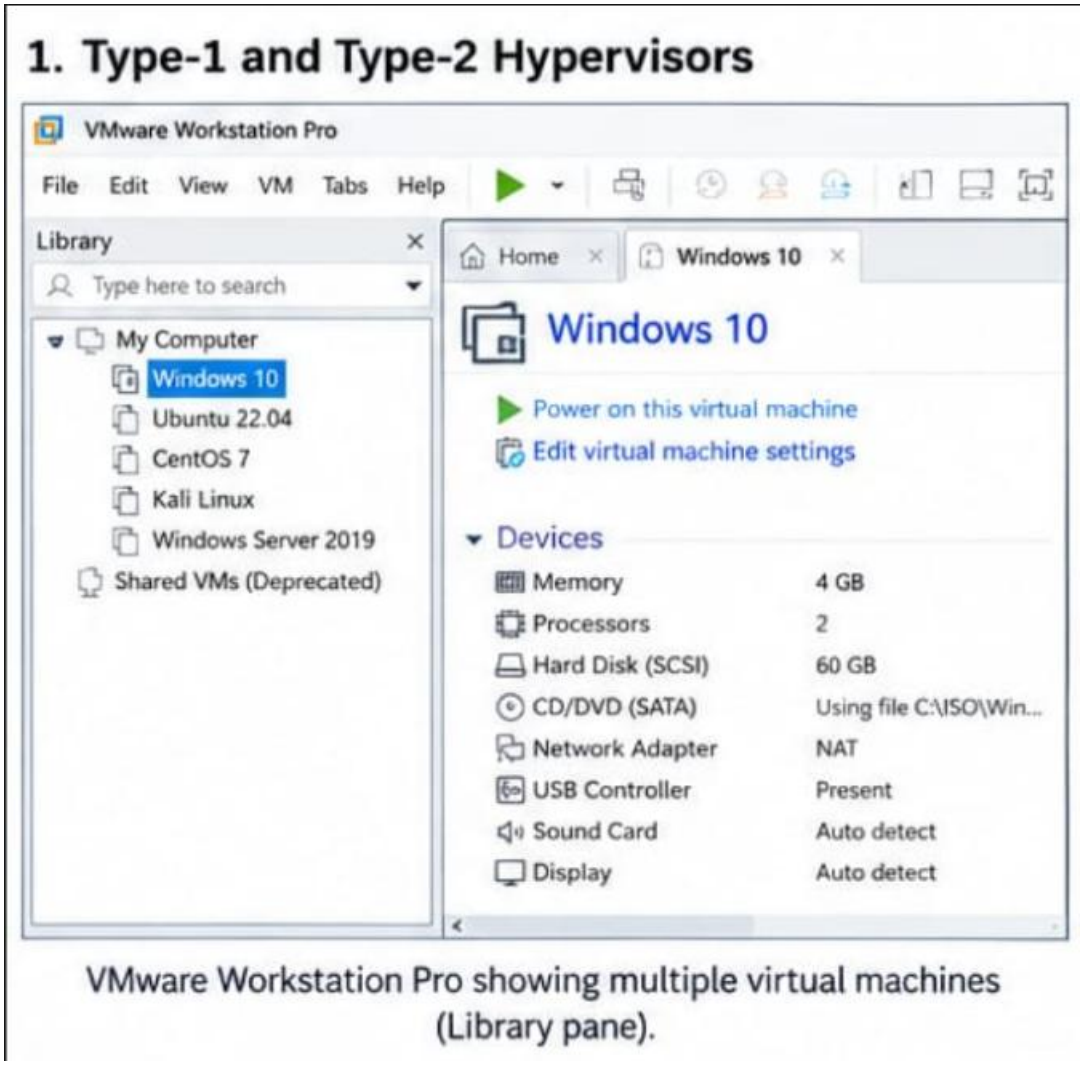
Question 1: Type-1 vs Type-2 Hypervisors for Mission-Critical Cloud Datacenters

A hypervisor abstracts physical hardware into virtual machines. A Type-1 (bare-metal) hypervisor, such as VMware ESXi, Microsoft Hyper-V, Xen or KVM, runs directly on the server hardware. A Type-2 (hosted) hypervisor, such as VMware Workstation or VirtualBox, runs as an application on top of a general-purpose host operating system.

Criteria	Type-1 (Bare-Metal)	Type-2 (Hosted)
Performance	Direct access to CPU, memory and I/O with hardware-assisted virtualization (Intel VT-x/AMD-V). Very low overhead and predictable latency.	Every request passes through the host OS, adding an extra layer, scheduling contention and higher latency. Performance varies with host load.
Security	Small attack surface (thin, purpose-built kernel). No general-purpose OS, browsers or user apps. Strong VM isolation.	Large attack surface: host OS vulnerabilities, drivers and other installed software can compromise all VMs.
Manageability	Centralized management (vCenter, SCVMM, OpenStack), live migration, HA, DRS, clustering, automation via APIs.	Managed per machine by hand. Limited clustering, HA and automation; not designed for large scale.
Scalability & Reliability	Supports hundreds of VMs per host and cluster-wide failover.	Limited by host OS resources; host crash brings down all VMs.
Typical Use	Production datacenters and public/private clouds.	Developer testing, labs, training, desktops.

Justification: A Type-1 hypervisor is the better choice for mission-critical applications. It offers near-native performance and consistent latency, a minimal and hardened attack surface, and enterprise features such as high availability, live migration and automated resource scheduling that keep services running during hardware failures or maintenance. Type-2 hypervisors are suitable only for development and testing.

Screenshot (Q1) – VMware Workstation Pro: Library pane showing multiple VMs (instances/regions)



VMware Workstation Pro showing multiple virtual machines (Library pane).

Question 2: Workload Isolation for Finance, Healthcare and Student Portals

How virtualization architecture supports isolation

- **Compute isolation:** Each workload runs in its own VM with a separate guest OS. The hypervisor uses hardware virtualization (VT-x/AMD-V, EPT/NPT, IOMMU) so that one VM cannot read or write another VM's memory or CPU state.
- **Resource isolation:** CPU shares, reservations, limits and memory quotas prevent a noisy student portal from starving the finance or healthcare systems.
- **Network isolation:** Separate virtual switches, VLAN/VXLAN segments, virtual firewalls and micro-segmentation keep traffic of each tenant apart.
- **Storage isolation:** Separate datastores/LUNs, LUN masking, per-tenant encryption keys and encrypted virtual disks protect data at rest.

- **Policy and compliance separation:** Dedicated clusters or host groups (affinity rules) can be used for regulated workloads (PCI-DSS for finance, HIPAA for healthcare), with role-based access control and audit logging.

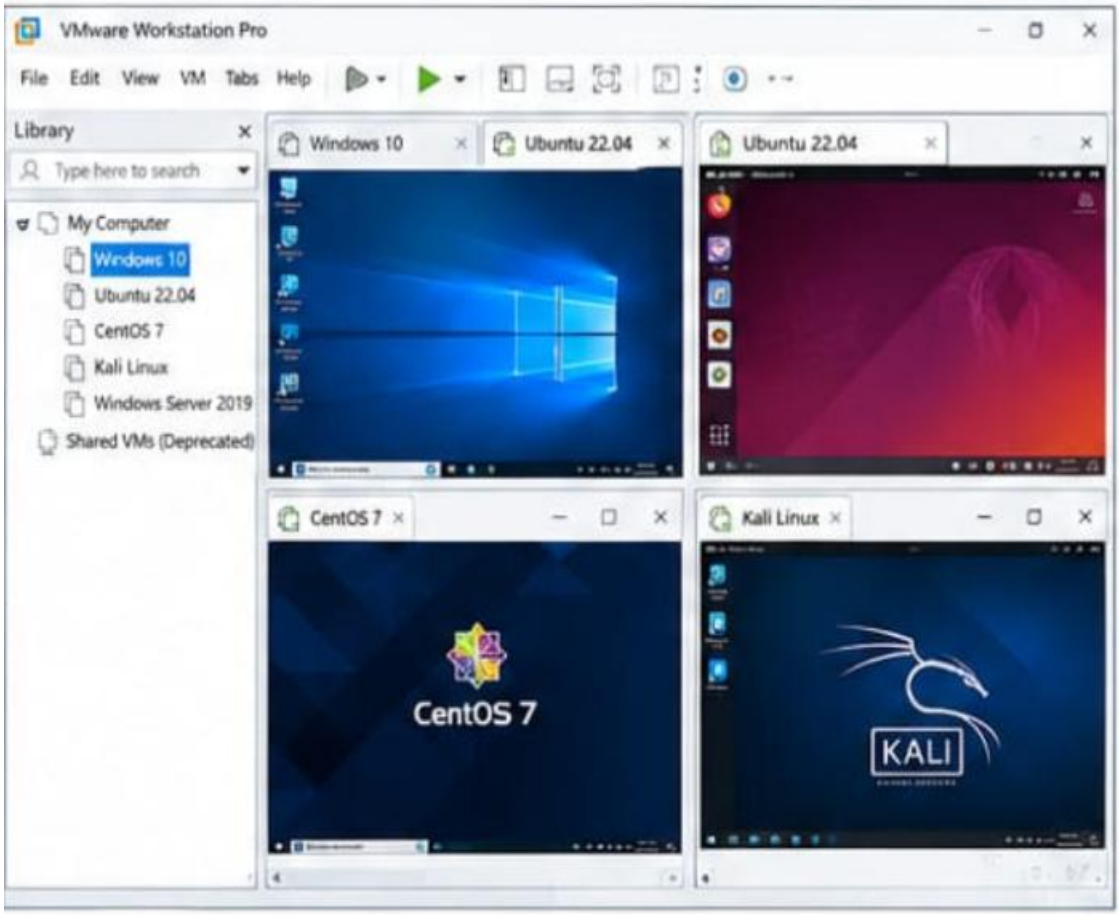
Two major attack surfaces

1. Hypervisor and shared hardware layer. Flaws in the hypervisor, device emulation or hypercalls can allow a VM escape, letting an attacker in one VM (for example the student portal) reach the host and other tenants. Shared CPU caches and memory also allow side-channel attacks (Spectre/Meltdown-type) that leak data between VMs. **Mitigation:** patch the hypervisor regularly, disable unused virtual devices, use IOMMU and hardware-based memory encryption, avoid co-locating high-sensitivity VMs with low-trust VMs, and apply CPU microcode mitigations.

2. Virtual network and management plane. Virtual switches, management consoles and APIs (vCenter, hypervisor admin interfaces) are high-value targets. Misconfigured VLANs, VM-to-VM traffic that bypasses firewalls, stolen administrator credentials or exposed APIs can expose all tenants at once. **Mitigation:** isolate the management network, enforce MFA and RBAC with least privilege, use micro-segmentation and distributed firewalls, encrypt management traffic and log all administrative actions.

Screenshot (Q2) – VMware Workstation Pro: Edit → Virtual Network Editor (separate NAT / Host-only networks f

2. Workload Isolation and Security



Different virtual machines running separate operating systems, showing isolation of workloads.

or isolation)

Question 3: VM Host Utilization Analysis

Snapshot: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s.

Metric	Observation	Likely Bottleneck / Meaning
CPU 92%	Far above the recommended 70–80% steady-state ceiling.	CPU over-commitment (too many vCPUs per physical core); VMs queue for CPU (high CPU-ready time), causing slow response.
Memory 88%	Close to the limit, little headroom for spikes.	Memory pressure may trigger ballooning, host swapping or compression, which pushes extra load onto the disk.

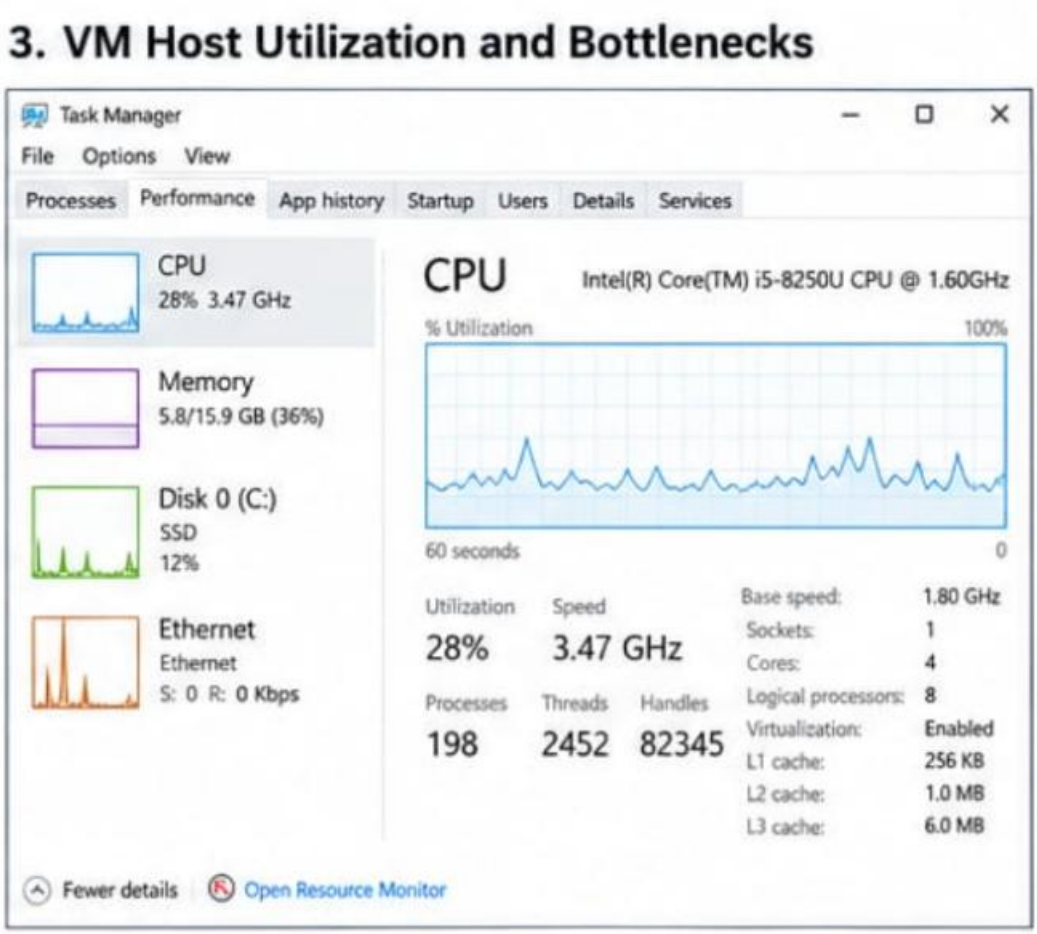
Metric	Observation	Likely Bottleneck / Meaning
Storage latency 35 ms	Healthy storage is typically below 10–20 ms.	Storage I/O contention: slow disks (HDD), saturated datastore, or too many VMs sharing one LUN; worsened by swapping.
Boot time 170 s	A normal VM boots in roughly 30–60 s.	A 'boot storm': many VMs reading disk images at once while CPU, memory and storage are already saturated.

Root cause: The primary bottleneck is storage I/O, combined with CPU and memory over-commitment. The three resources are linked: memory pressure causes swapping, which raises disk latency, which slows boot and application response, while high CPU contention delays VM scheduling.

Corrective actions

- **Load balancing:** Use live migration (vMotion) and DRS to move VMs to under-used hosts; add hosts to the cluster if all are busy.
- **CPU:** Right-size VMs (reduce unnecessarily large vCPU counts), lower the vCPU-to-core ratio, and set resource shares/limits.
- **Memory:** Reduce over-allocation, reclaim idle memory, add RAM, and enable memory-efficient techniques (page sharing, ballooning) only with sufficient headroom.
- **Storage:** Move to SSD/NVMe or tiered storage, spread VMs across multiple datastores, enable caching, and apply storage I/O control/QoS.
- **Boot time:** Stagger VM startups, use templates and linked clones, and place boot images on fast storage.
- **Monitoring:** Set alerts (CPU > 80%, latency > 20 ms) and review trends to plan capacity ahead of demand.

Screenshot (Q3) – VMware Workstation Pro: VM → Manage → Clone (replica/scale-out to spread load)



Question 4: Software Defined Datacenter (SDDC) vs Traditional Datacenter

In an SDDC, compute, storage, networking and security are all virtualized, pooled and controlled by software through policies and APIs, instead of being tied to specific hardware devices configured manually. This makes the infrastructure programmable, automated and fast to change.

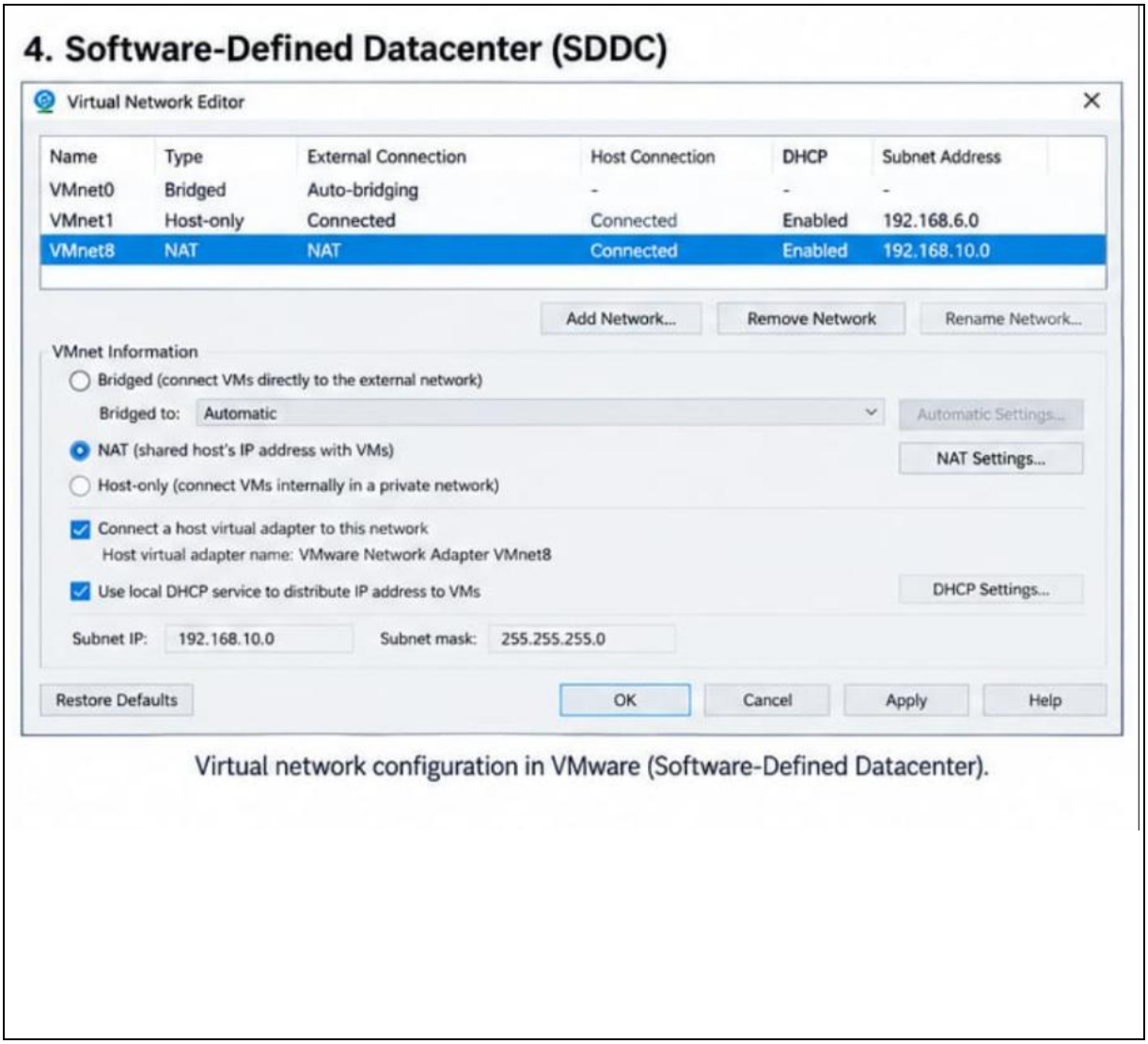
Aspect	Traditional Datacenter	Software Defined Datacenter
Provisioning	Days or weeks: hardware ordering, cabling, manual configuration.	Minutes: self-service portals and automated templates.
Compute	Dedicated physical servers per application; low utilization.	Server virtualization pools CPU/RAM; VMs/containers created, cloned and migrated on demand.
Storage	Separate SAN/NAS silos managed per array.	Software-defined storage (vSAN, Ceph) pools disks into policy-driven volumes that scale out.
Network	Hardware switches/firewalls configured device by device.	Network virtualization (SDN, VXLAN, NSX) with centralized control plane and programmable networks.
Scaling	Manual, slow, over-provisioned.	Elastic, automatic scale-up/out.
Management	Many separate tools; human error prone.	Single management/orchestration layer with policy-based automation.
Cost model	High CAPEX, vendor lock-in.	Commodity hardware, lower OPEX, better utilization.

Why agility improves

- **Compute virtualization:** New VMs are deployed from templates in minutes, resources can be resized, and VMs can be live-migrated for maintenance without downtime.
- **Storage virtualization:** Storage is allocated by policy (performance, replication, capacity) automatically; thin provisioning, snapshots and cloning enable rapid backup and test environments.
- **Network virtualization:** Virtual networks, load balancers and firewall rules are created through software, without rewiring, and follow the VM when it moves; micro-segmentation adds security per workload.

Conclusion: Because all three layers are abstracted and automated through one orchestration layer, an SDDC lets organizations respond quickly to changing business demand, scale elastically, reduce human error and cost, and achieve faster recovery, which a traditional hardware-centric datacenter cannot match.

Screenshot (Q4) – VMware Workstation Pro: VM → Snapshot → Take Snapshot / Snapshot Manager (rapid versioning and recovery)



Virtual network configuration in VMware (Software-Defined Datacenter).

Question 5: Identity Mismatch in a Multi-Cloud Federation

Analysis of the problem

In a multi-cloud federation each provider has its own identity system (for example IAM, Azure AD/Entra ID and a private-cloud LDAP). The project fails because these systems are not trusted or aligned with one another. Typical causes are:

- **Different identity formats and naming (user IDs, email, UPN), so the same person has different identities in each cloud.**
- **Different protocols and token formats (SAML, OAuth 2.0/OIDC, Kerberos, proprietary tokens) with no common trust agreement.**
- **Inconsistent attributes, roles and permission models, so authorization decisions differ between clouds.**

- **No shared trust relationship, certificate exchange or key management between providers.**
- **Unsynchronized account lifecycle (joiners/leavers), causing orphan accounts and security gaps, plus privacy risk from sharing too many attributes.**

Proposed secure identity and privacy design

1. Federated identity with a trusted Identity Provider (IdP). Use one authoritative IdP (or an identity broker) per organization. Each cloud acts as a Service Provider (SP) and trusts the IdP through a signed metadata exchange, enabling Single Sign-On using SAML 2.0 or OpenID Connect/OAuth 2.0.

2. Common identity and attribute mapping. Define a global unique identifier (federated ID) and a standard claim schema. Map local attributes and roles to it at the broker so that every provider interprets the identity the same way.

3. Standard provisioning. Use SCIM for automated account creation, update and removal so that leaver accounts are deprovisioned across all clouds quickly.

4. Strong authentication and authorization. Enforce multi-factor authentication, role-based and attribute-based access control (RBAC/ABAC) with least privilege, and short-lived signed tokens with limited scope.

5. Privacy protection. Release only minimum necessary attributes to each provider, use pseudonymous/pairwise identifiers so users cannot be tracked across clouds, obtain user consent, and comply with data-protection rules (GDPR, HIPAA).

6. Security and trust management. Encrypt all traffic with TLS, sign and encrypt assertions, manage and rotate certificates and keys, and apply centralized audit logging and monitoring across all providers. Define a trust framework (SLAs, policies, identity assurance levels) among the federation members.

Result: Users sign in once with a consistent, verifiable identity; each cloud makes consistent authorization decisions; and personal data is protected, which resolves the identity mismatch and secures the federation.

Screenshot (Q5) – VMware Workstation Pro: Edit → Virtual Network Editor (Bridged/NAT/Host-only = hybrid or multi-cloud connectivity)

